



SCS 1306 - Linear Algebra

Tutorial 01

01) Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 7 & 8 & 9 \end{bmatrix}$, $B = \begin{bmatrix} 5 & 6 & 7 \\ 3 & 4 & 5 \end{bmatrix}$ and $C = \begin{bmatrix} -1 & 5 & 7 \\ 5 & -6 & 5 \end{bmatrix}$.
Find each of the following:

(i) $A + B$,

(ii) $A - B$,

(iii) $A + B - C$,

(iv) $2A - B - 3C$.

02) Let $A = \begin{bmatrix} -5 & 2 & 0 \\ 7 & -3 & 4 \\ -1 & 3 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 5 & -1 & 8 \\ 6 & -14 & 2 \\ 9 & 5 & 1 \end{bmatrix}$.
Evaluate each of the following:

(i) A^T and B^T ,

(ii) $(A + B)^T$,

(iii) $A^T + B^T$,

(iv) $(AB)^T$,

(v) $(-A)^T$,

(vi) $(2A - 5BA)^T$,

(vii) $\text{tr} A + B$,

(viii) $\text{tr}(2A - 5BA)^T$,

(ix) $\text{tr}(AB)$,

(x) $\text{tr}(AB)^T$.

03) Verify the result $A(B + C) = AB + AC$ by using the following 3 matrices:

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 1 & 0 & 1 \end{bmatrix},$$

$$B = \begin{bmatrix} 2 & -1 & -1 \\ 1 & 0 & -1 \\ -2 & 1 & 2 \end{bmatrix},$$

$$C = \begin{bmatrix} 3 & 1 & 4 \\ 2 & 1 & 3 \\ 2 & 2 & 1 \end{bmatrix}.$$

04) (a) Solve the following matrix equation for x, y , and z :

$$\begin{bmatrix} x & 3 \\ z & 2 \end{bmatrix} + \begin{bmatrix} 2 - y & z \\ 2 - z & -x \end{bmatrix} = \begin{bmatrix} 3 & 7 \\ 2 & 0 \end{bmatrix}.$$

(b) Let $A = \begin{bmatrix} 3 & 4 \\ -1 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 2 \\ -1 & 2 \end{bmatrix}$.

Find the matrix X satisfying the matrix equation $2X + B = 3A$.

(c) Find a solution $X \in \mathbb{M}_{2 \times 2}(\mathbb{R})$ for $X^2 - 2X = \begin{bmatrix} -1 & 0 \\ 6 & 3 \end{bmatrix}$.